ELIXIR biovalidator

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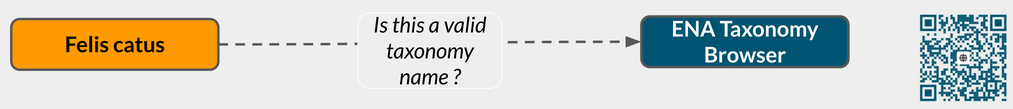
Introduction

To improve the FAIRness of ever increasing amounts of complex biomedical data, validation of both the structure and the content of data is required. The ELIXIR biovalidator[1] allows to verify syntactic and semantic compliance of JSON documents by extending the existing JSON Schema syntax.
The ELIXIR biovalidator is an extension of widely used JSON Schema validation library, AJV[2] to support validation use cases important for life sciences such as ontology, taxonomy and identifier validation. The ELIXIR biovalidator can be run as a service or a one time command. The service provide an API for programmatic users and a web based UI for ad hoc use.

TAXONOMY VALIDATION

`isValidTaxonomy`

Checks if the given taxonomy exists in European Nucleotide Archive's (ENA)[3] Taxonomy Browser*.

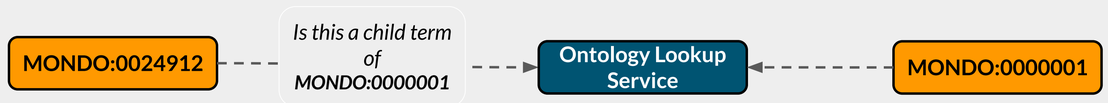


ONTOLOGY VALIDATION

`graphRestriction` `isChildTermOf` `isValidTerm`

Validates provided ontologies using EMBL-EBI Ontology Lookup Service (OLS)[4].

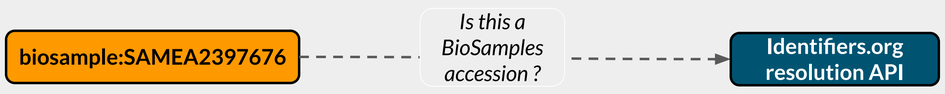
- "isValidTerm" evaluates if provided term is an ontology.
- "isChildTermOf" evaluates if the term provided is a child of a term given in the schema.
- "graphRestriction" allows to do more complex validation using rdfs syntax.



IDENTIFIER VALIDATION

`isValidIdentifier`

Evaluates if a given identifier has a correct format using identifiers.org[5] resolution API. Checks if given identifier belong to the list of allowable identifier types in the schema.

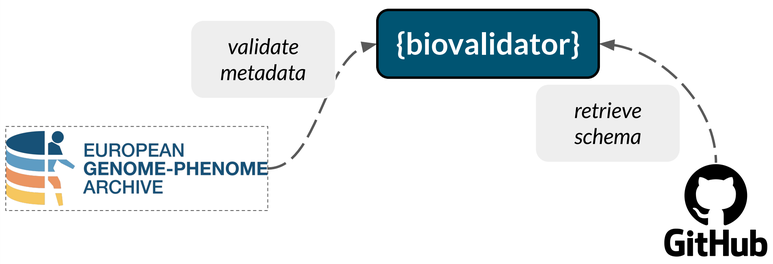


FUTURE

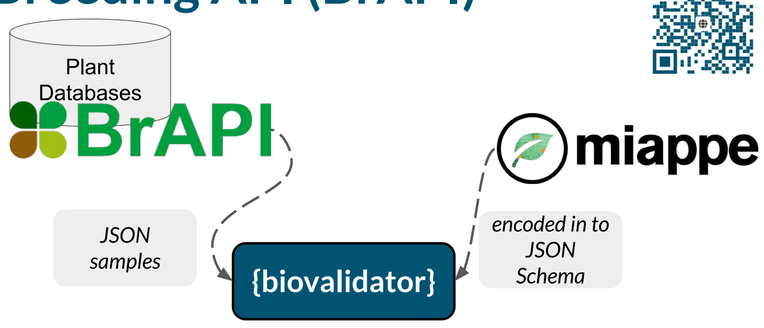
- Graph validation - dependency validation use cases not supported by vanilla JSON Schema
- Improvements to cache management
- Improvements to observability

USERS

EGA European Genome-Phenome Archive (EGA)

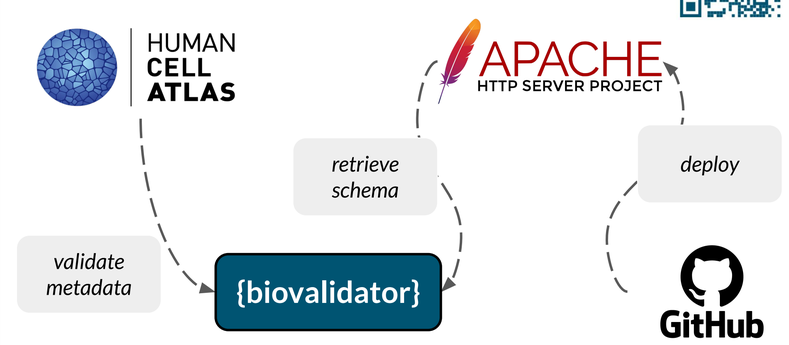


Breeding API (BrAPI)

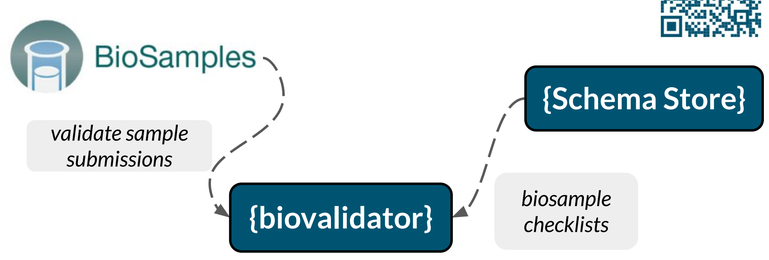


MIAPPE - Minimum Information About Plant Phenotyping Experiments

Human Cell Atlas (HCA)



EBI - BioSamples Database



*JSON Schema Store is a Java application developed at EBI BioSamples to store versioned JSON Schema.

IMPLEMENTATION

The ELIXIR biovalidator can be run as a server or a one time Command Line Interface (CLI) command.

Service

`fast` `caching` `ui` `api` `external users`

```
> node ./src/biovalidator.js
```

CLI Command

`convenient` `local machine`

```
> node ./src/biovalidator.js \
  --data=test_data.json --schema=test_schema.json
```

Docker

`development` `integration`

```
> docker run -p 3020:3020 -d
quay.io/ebi-ait/biovalidator:2.0.0
```

elixir-europe/biovalidator

Contact
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References

1. Liyanage, I., Burdett, T., Droesbeke, B., Erdos, K., Fernandez, R., Gray, A., ... & Coppens, F. (2022). ELIXIR biovalidator for semantic validation of life science metadata. Bioinformatics, 38(11), 3141-3142.
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